

Braid Presentations for Concurrent Programs

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Abstract

Ziemiański's cube chain category $\text{Ch}(K)$ is a combinatorial model for the space of directed paths on a precubical set K . Paliga and Ziemiański built a braid invariant for precubical sets by observing a relation between $\pi_1|\text{Ch}(Z)|$ and braids, where Z is the terminal precubical set. However, that invariant trivializes for Euclidean K because $|\text{Ch}(I^n)|$ is contractible [PZ21, Prop. 6.1]. This paper constructs a localization of $\text{Ch}(K)$ at morphisms that preserve the order of coordinates, called $\text{Br}(K)$. And we will show that this structure is the source of the braid structure of $\pi_1|\text{Ch}(Z)|$. But, there is no nerve or geometric realization in this technique. Instead we build a 2-polygraph giving an artin braid-like presentation of $\text{Br}(K)$. The $\{0, 1, 2\}$ -cells of the polygraph are the dimension $\{0, 1, 2\}$ chains of K . Intuitively, 3-cells of K contribute braid relations, and pairs of non-intersecting 2-cells of K contribute commuting relations. This is not just a metaphor: at $K = Z$, we get the artin relations and generators exactly. So $\text{Br}(Z)$ is equivalent to the positive braid category. The presentation is natural in K , providing a generalization of the existing work. As an application, we establish a link to the hyperplane arrangement literature. The way braids are constructed via Salvetti's theorem for the braid arrangement exactly matches the polygraph we build here for the cube. All this work is formalized in Lean [Sto26].

1 Introduction

Ziemiański's cube chain category $\text{Ch}(K)$ is a combinatorial model for the executions of a bipointed precubical set K [Zie20, Thm. 7.6]. Paliga and Ziemiański established a homotopy equivalence between $|\text{Ch}(Z)|$ and configuration spaces of points in the plane, where Z is the terminal precubical set. It extends, by naturality, to an invariant of any precubical set K . However, it trivializes on any Euclidean complex, see [PZ21, Prop. 6.1]. The proof is one line: the invariant factors through $|\text{Ch}(I^n)|$, which is contractible.

This paper explains that the braid structure is present in $\text{Ch}(K)$ without passing through any homotopy equivalences. We will construct a set of morphisms, W , in $\text{Ch}(K)$. And define the localization $\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}]$. Then we will build a 2-polygraph giving an artin-like presentation for $\text{Br}(K)$ in terms of chains of K . We do not pass via a homotopy equivalence to the path space, as it gives us several advantages. Firstly, we preserve a lot of the directed structure of in K , so $\text{Br}(I^n)$ it does not collapse. Secondly, it permits us to stay with positive braids which yields a simpler theory.

The the cells of our polygraph are straightforward. A chain of K has a list of positive integers for the dimension of the cubes. Let the degree of a chain be the sum of the list minus the length. Then the polygraph's n -cells are the degree n chains, for $n \in \{0, 1, 2\}$. The attachements of the 2 cells to the 1 cells will take some work to formalize, but the following pictures gives a clear intuition about

the attachements. The key point is that every degree 2 chain has one of these forms.

► *Read Figure 1 and Figure 2 for the reader: a run is a way of traversing the blocks of a chain; a degree-one chain has a single square, hence exactly two runs, and it is the generator that flips that square; a degree-two chain offers two flips, so its runs form a polygon with two routes, and the 2-cell says those routes agree. Say that these two pictures are the only shapes this can take, which is why the relations come out Artin's.*

Theorem 1.1 (Theorem 5.3, Theorem 6.1, Theorem 7.5). *Let K be a bipointed precubical set, $\text{Br}(K)$ the localization above (Definition 3.5) and $\text{Art}(K)$ the 2-polygraph just described (Definition 4.13), presenting $\langle \text{Art}(K) \rangle$.*

1. *There is an equivalence $E_K : \langle \text{Art}(K) \rangle \rightarrow \text{Br}(K)$.*
2. *Both constructions are functorial in K , and E is natural: for every $f : K \rightarrow L$,*

$$E_L \circ \langle \text{Art}(f) \rangle = \text{Br}(f) \circ E_K.$$

3. *$\text{Art}(Z)$ is isomorphic to Artin's braid polygraph (Definition 7.4). Consequently $\text{Br}(Z)$ is equivalent to the opposite of the positive braid category.*

None of the three assumes anything about K .

► *Say why $\text{Art}(K)$ deserves the name “Artin polygraph” before (iii) is proved: its 2-cells already split into the two species of Proposition 4.12, a hexagon at a block of dimension 3 and a square at two blocks of dimension 2, which are Artin's braid and commutation relations. Then note that (iii) makes this literal.*

► *Relation to Gaussent–Guiraud–Malbos [GGM15]: the same 2-cells, there from Garside theory and here from the geometry of K , with the Artin case the value at a point.*

Formalisation. All of the below is formalised in Lean 4 on mathlib [Sto26], sorry-free. The objects $\text{Br}(K)$, W and $\text{Art}(K)$ are `ChainCat.Br`, `ChainCat.W` and `Paper.poly`; the three parts of Theorem 1.1 are `Paper.paperPresents`, `Paper.paperSquare` and `Paper.paperArtinIso`. Each statement below appears in `CubeChains/Paper.lean` as an `example` discharged by one citation.

Conventions. We assume the reader knows [PZ21; Zie20] and use its notation: $\square_*$ is the category of bipointed precubical sets, with Z final and I^n the n -cube. Put $[n] := \{1, \dots, n\}$. Composition is classical: gf means “first f , then g ”. A word in a polygraph is a path read left to right, so $\sigma_i \sigma_k$ is the arrow $\sigma_k \circ \sigma_i$.

► *Explain the two $^{\text{op}}$ s: refinements run from finer to coarser [PZ21; Zie20], and words read left to right, so $\text{Br}(K)$ localizes $\text{Ch}(K)^{\text{op}}$ and $\langle \mathcal{A} \rangle$ is $\mathbf{Braid}^{+, \text{op}}$.*

2 Background

► *One paragraph placing Proposition 2.1: everything downstream is bookkeeping about the events of a wedge, and this is the only place the precubical structure is touched.*

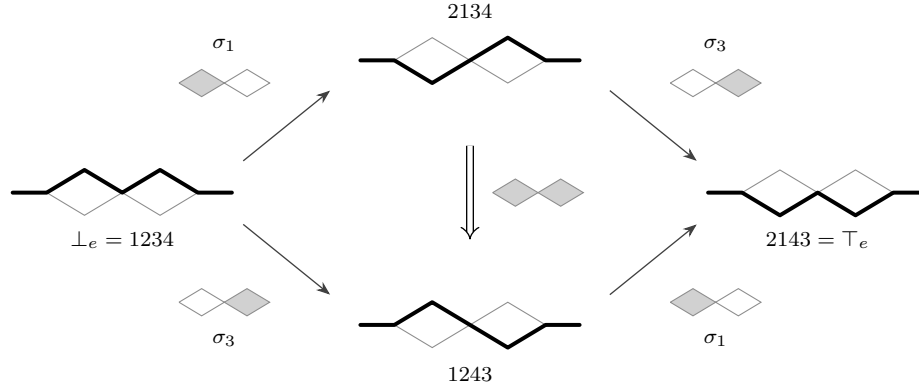


Figure 1: The four runs over the degree-two chain $e = V[2, 2]$. Each arrow is decorated with the square that witnesses the reordering. And each node is decorated with the degree 0 refinement's permutation. So the 2 cell attaches $\sigma_1 \sigma_3 \rightarrow \sigma_3 \sigma_1$.

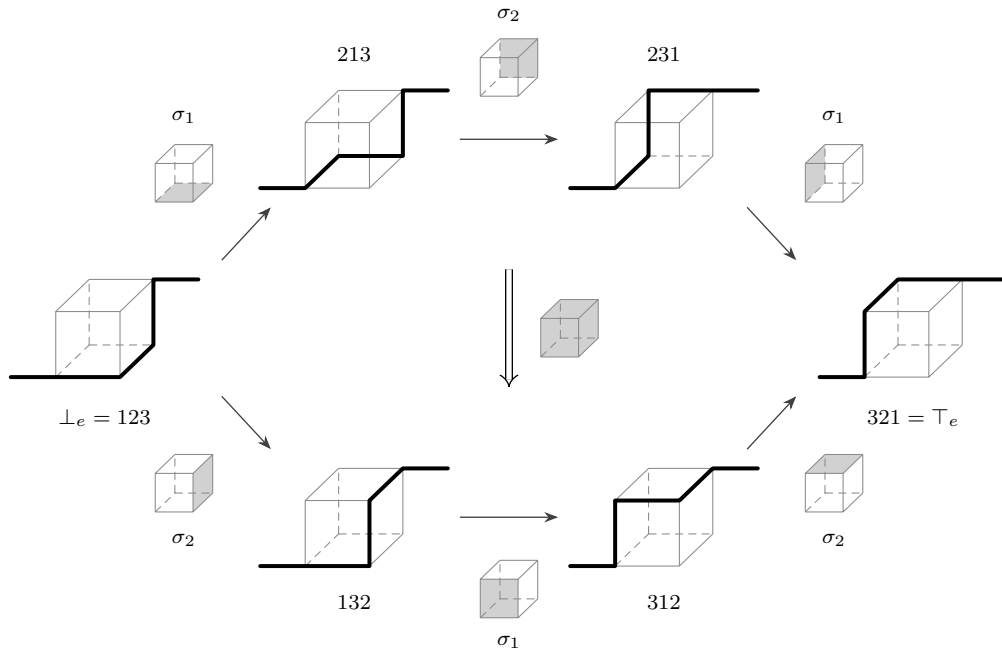


Figure 2: The six runs over the degree-two chain $e = V[3]$. Again each arrow is decorated with the square that witnesses the reordering. And each node is decorated with the degree 0 refinement's permutation. So we get the attachment $\sigma_1 \sigma_2 \sigma_1 \rightarrow \sigma_2 \sigma_1 \sigma_2$.

Cube chains. A **shape** is a list $a = [a_1, \dots, a_r]$ of positive integers; it names the serial wedge $\bigvee a := I^{a_1} \vee \dots \vee I^{a_r}$, whose **blocks** are the I^{a_i} , and we put $|a| := \sum_i a_i$ and $\deg a := |a| - r$. Objects of $\text{Ch}(K)$ are the bipointed maps $\bigvee a \rightarrow K$ and its morphisms, the refinements, the maps of wedges over K . We write $|d|$ and $\deg d$ for the event count and degree of a chain's shape, and $\text{Ch}(u)$ for the pushforward along $u : K \rightarrow L$, which postcomposes the classifying map and leaves the shape alone. A chain of degree 0 is a **run**, and the subcategory of runs is discrete, a refinement losing no block being an identity.

We import three facts from [PZ21, §2]: serial wedges are non-self-linked and admit an altitude, for which the final vertex of $\bigvee a$ has altitude $|a|$; a bipointed map between connected objects admitting altitude preserves it [PZ21, Prop. 2.7]; and concatenation $\text{Ch}(P) \times \text{Ch}(Q) \rightarrow \text{Ch}(P \vee Q)$ is an isomorphism when $P \vee Q$ is non-self-linked and admits altitude [PZ21, Lem. 2.11].

The events of a shape. Send I^n to its set of directions $[n]$, and a face embedding to the monotone injection naming the directions it leaves free. The left Kan extension of that functor along Yoneda is cocontinuous and vanishes at I^0 , so the pushout defining a wedge becomes a coproduct: the **events** of $\bigvee a$ are the pairs (i, c) with $i \in [r]$ and $c \in [a_i]$, one for each direction of each block. Order them lexicographically and put $\text{blk}_a(i, c) := i$, so the blocks of a are the fibres of blk_a and are intervals for that order; we drop the subscript when the shape is clear. Numbering the events $1, \dots, N$ in order identifies them with $[N]$, $N = |a|$, and we use whichever description is convenient. Write f_* for the map on events induced by a map f of serial wedges.

Proposition 2.1. *Let $f : \bigvee a \rightarrow \bigvee b$ in $\square_*$, with a and b shapes on N and M events. Then $N = M$; the map f_* is a permutation of $[N]$, and $f \mapsto f_*$ is injective; and there is a monotone $\text{blk}(f)$ with*

$$\text{blk}_b \circ f_* = \text{blk}(f) \circ \text{blk}_a, \quad (1)$$

the restriction of f_ to each block of a being increasing.*

Proof. By the pushout, f is determined by its restrictions to the blocks of a , each of which lands in a single block of b because $a_i > 0$; that gives $\text{blk}(f)$, equation (1) and the last clause, and a face embedding is determined by the directions it leaves free, so f_* determines f . That f_* is injective and $\text{blk}(f)$ monotone is an induction on the length of b : by [PZ21, Lem. 2.11] a map into $I^m \vee \bigvee b'$ splits over a splitting of a , and at each glued vertex the two halves have disjoint images, since the coordinates a face embedding leaves free before the vertex are already 1 there. Finally f preserves altitude [PZ21, Prop. 2.7], so $N = M$ and the injection f_* is a permutation. $\square$

Presented categories. A **2-polygraph** $\mathcal{P}$ consists of a set V of 0-cells, families $\mathcal{P}_1(x, y)$ and $\mathcal{P}_2(x, y)$ for $x, y \in V$, and two maps $\text{src}, \text{tgt} : \mathcal{P}_2(x, y) \rightarrow \{\text{paths } x \rightarrow y\}$ into the free category on the 1-cells; $\text{src}(\alpha)$ and $\text{tgt}(\alpha)$ are the **words** of α . A morphism of polygraphs is a map in each dimension commuting with src and tgt on the nose, and **Poly** is the resulting category. The category $\langle \mathcal{P} \rangle$ **presented by** $\mathcal{P}$ is the free category on the 1-cells modulo the congruence generated by $\text{src}(\alpha) = \text{tgt}(\alpha)$, with quotient functor q .

Proposition 2.2 ([Ara+23, Ch. 2]). *Let $\mathcal{P}$ be a 2-polygraph and C a category.*

1. *To give a functor $\langle \mathcal{P} \rangle \rightarrow C$ is to give an object of C for each 0-cell and an arrow of C for each 1-cell, subject to the two words of every 2-cell being sent to the same arrow.*
2. *Two functors out of $\langle \mathcal{P} \rangle$ that agree on 0-cells and 1-cells are equal.*
3. *A morphism of polygraphs bijective in every dimension is an isomorphism.*

Proposition 2.3 ([GZ67, Ch. I]). *Let W be a class of morphisms of a category C . There is a functor $Q : C \rightarrow C[W^{-1}]$ inverting W through which every functor inverting W factors uniquely, and two functors out of $C[W^{-1}]$ that agree after Q are equal.*

► *One sentence: the only arrows of $\text{Br}(K)$ written down below are composites $Q(g^{\text{op}})Q(m^{\text{op}})^{-1}$ coming from a cospan whose first leg is order preserving, and every identity used about them is an instance of Q being a functor.*

Newman’s lemma. Write $\text{Hasse}(P)$ for the quiver of covers of a poset P . A map of quivers $\text{Hasse}(P) \rightarrow C$ into a category extends to paths by composition, and we read it on paths without further comment.

Theorem 2.4 (coherent Newman’s lemma; [Hue80], [Ara+23, Ch. 4]). *Let P be a finite poset and $\varphi : \text{Hasse}(P) \rightarrow C$ a map of quivers. Suppose that for every pair of distinct covers $x \triangleleft y_1$, $x \triangleleft y_2$ and every common upper bound z of y_1 and y_2 there are $j \leq z$ and paths p_i from y_i to j with $\varphi(x \triangleleft y_1) \varphi(p_1) = \varphi(x \triangleleft y_2) \varphi(p_2)$. Then φ sends parallel paths to equal arrows, so it extends uniquely to a functor $P \rightarrow C$; and a functor out of P is determined by its values on covers.*

3 Squares, crossings, and the braiding

► *One paragraph: a square is a pair of events of one block, and the whole construction runs on the question of which squares a map reverses.*

Throughout, a and b are shapes on N events.

Definition 3.1. A **square** of a is a pair $s = (p, q)$ of events with $p < q$ and $\text{blk}(p) = \text{blk}(q)$; write $\text{Sq}(a)$ for the set of them, so $|\text{Sq}(a)| = \sum_i \binom{a_i}{2}$. A set $R \subseteq \text{Sq}(a)$ orients the complete graph on each block: send (p, q) to the edge $q \rightarrow p$ when $(p, q) \in R$, and to $p \rightarrow q$ otherwise. Call R **biclosed** when no block’s tournament has a directed 3-cycle. On a triple $p < q < r$ of one block the two cycles are $p \rightarrow r \rightarrow q \rightarrow p$ and $p \rightarrow q \rightarrow r \rightarrow p$, so this says

$$(p, q) \in R \text{ and } (q, r) \in R \implies (p, r) \in R, \quad (p, r) \in R \implies (p, q) \in R \text{ or } (q, r) \in R.$$

We write $\text{Sq}(d)$ for the squares of a chain’s shape, and f_* also for the induced injection $\text{Sq}(a) \rightarrow \text{Sq}(b)$, $(p, q) \mapsto (f_*p, f_*q)$, well defined because f_* is increasing on blocks (Proposition 2.1).

Definition 3.2. The **crossings** of a map of serial wedges $f : \bigvee a \rightarrow \bigvee b$ are the inversions of the relabelling it performs:

$$\text{Cr}(f) := \{ (p, q) : p < q \text{ and } f_*^{-1}(q) < f_*^{-1}(p) \}.$$

Lemma 3.3. *The map f_*^{-1} preserves the order of any two events lying in distinct blocks of b . Hence $\text{Cr}(f) \subseteq \text{Sq}(b)$, the two events of a crossing lying in one block of b . A square (p, q) of b lies in the image of f_* if and only if $f_*^{-1}p$ and $f_*^{-1}q$ lie in one block of a .*

Proof. Let $u < v$ lie in distinct blocks of b , so $\text{blk}_b(u) < \text{blk}_b(v)$, and put $w = f_*^{-1}u$, $w' = f_*^{-1}v$. Then (1) gives $\text{blk}(f)(\text{blk}_a w) < \text{blk}(f)(\text{blk}_a w')$, so $\text{blk}_a(w) < \text{blk}_a(w')$ as $\text{blk}(f)$ is monotone, so $w < w'$ as blocks are intervals. An inversion of f_*^{-1} therefore has both events in one block of b . For the last claim, a square in the image pulls back to a square, whose events share a block; conversely if the two preimages share a block then, f_* being increasing there, they form a square mapping to (p, q) . $\square$

Lemma 3.4. For $f : \bigvee a \rightarrow \bigvee b$ and $g : \bigvee b \rightarrow \bigvee c$,

$$\text{Cr}(gf) = g_*(\text{Cr } f) \sqcup \text{Cr}(g),$$

a disjoint union, and $\text{Cr}(\text{id}) = \emptyset$.

Proof. Let s be a square of c . If $s = g_*(t)$ then $s \notin \text{Cr}(g)$, the two events of t sharing a block (Lemma 3.3) — which also gives disjointness — and $(gf)_*^{-1}$ agrees with f_*^{-1} on the events of t , so $s \in \text{Cr}(gf)$ if and only if $t \in \text{Cr}(f)$. If s is not in the image of g_* then it is not in $g_*(\text{Cr } f)$, and its two events pull back along g_* into distinct blocks of b , so the second part of Lemma 3.3 applied to f gives $s \in \text{Cr}(gf)$ if and only if $s \in \text{Cr}(g)$. $\square$

Definition 3.5. A refinement f is **order preserving** when $\text{Cr}(f) = \emptyset$. Write $W = W_K$ for the class of order-preserving refinements of $\text{Ch}(K)$, and

$$\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}].$$

Proposition 3.6. A refinement f is order preserving if and only if $f_* = \text{id}_{[N]}$. Hence W contains the identities; $gf \in W$ if and only if $f \in W$ and $g \in W$; any two order-preserving refinements with the same source and target are equal; and $\text{Ch}(u)$ carries W_K into W_L , so it descends to $\text{Br}(u) : \text{Br}(K) \rightarrow \text{Br}(L)$, strictly functorially in u .

Proof. A permutation with no inversions is the identity, and $\text{Cr}(f)$ is the inversion set of f_*^{-1} , so $\text{Cr}(f) = \emptyset$ if and only if $f_*^{-1} = \text{id}$, that is $f_* = \text{id}$. The consequences follow: the second from Lemma 3.4 with g_* injective, the third because f_* determines f (Proposition 2.1), and the last because $\text{Ch}(u)$ changes neither shape nor wedge map, with the descent given by Proposition 2.3. $\square$

► One or two sentences reading Proposition 3.6 back: an order-preserving refinement is one that only glues adjacent blocks, the elementary ones being $\bigvee[\dots, p, q, \dots] \rightarrow \bigvee[\dots, p + q, \dots]$.

4 The Artin polygraph of K

► Frame the section: the runs over a chain form a poset; its covers are the degree-one chains over that chain, and the rank-two intervals are the degree-two ones. That is the whole polygraph.

Fix K , and let d be a chain of K on N events.

4.1 Runs over a chain

Definition 4.1. A **run over** d is a refinement $x : X \rightarrow d$ whose source is a run; write $\mathcal{R}(d)$ for the set of these.

Such a refinement is determined by its map of wedges, the classifying map of its source being forced, so $\mathcal{R}(d)$ is the set of runs of $\bigvee a$. That source has shape $[1, \dots, 1]$, so x_* enumerates the events of d ; write $p <_x q$ when x_* reaches p before q . Then $\text{Cr}(x)$ is the inversion set of $<_x$.

Lemma 4.2. The order $<_x$ takes the blocks of d in turn: if $\text{blk}(p) < \text{blk}(q)$ then $p <_x q$. Hence x amounts to a choice of linear order on each block, and blk is monotone for $<_x$ with interval fibres.

Proof. The blocks of the source are singletons, so $\text{blk} \circ x_* = \text{blk}(x)$, monotone by Proposition 2.1. A monotone surjection out of a chain has interval fibres, here the blocks of d ; the only freedom left is the order inside each. $\square$

Theorem 4.3. $x \mapsto \text{Cr}(x)$ is a bijection from $\mathcal{R}(d)$ onto the biclosed subsets of $\text{Sq}(d)$.

Proof. The map and the condition are both local to a block (Lemma 4.2), so fix one. A run restricts there to a linear order; its tournament is that order, which has no cycle at all, and Cr restricts to its inversion set. Conversely a biclosed R gives a tournament with no 3-cycle, hence a transitive one — if $p \rightarrow q$ and $q \rightarrow r$ then $r \rightarrow p$ would close a 3-cycle — and a transitive tournament is a linear order, whose inversion set is R . $\square$

► *Note that for a single block this is the classical description of inversion sets, and that inclusion of them is the right weak order [BB05, §3.1]; the order theory below is that statement, block by block.*

We order $\mathcal{R}(d)$ by $x \leq y : \iff \text{Cr}(x) \subseteq \text{Cr}(y)$, and write $\perp_d, \top_d$ for its least and greatest elements, which exist because $\emptyset$ and $\text{Sq}(d)$ are biclosed.

Proposition 4.4. Let $q : o \rightarrow d$ be a refinement. Composition with q is an order embedding $\mathcal{R}(o) \rightarrow \mathcal{R}(d)$, and

$$\text{Cr}(qx) = q_*(\text{Cr}(x)) \sqcup \text{Cr}(q).$$

Moreover $q \perp_o = \perp_d$ if and only if $q \in W$; for $x \in \mathcal{R}(d)$ the conditions $x \in W$, $\text{Cr}(x) = \emptyset$ and $x = \perp_d$ agree; $\perp_X = \text{id}_X$ for a run X ; and $\text{dom } \perp_d = \text{dom } \perp_c$ whenever $c \rightarrow d$ lies in W .

Proof. The formula is Lemma 3.4. Since q_* is injective with image disjoint from $\text{Cr}(q)$ (Lemma 3.3), $qx \leq qy$ if and only if $x \leq y$, and $\text{Cr}(q \perp_o) = \text{Cr}(q)$. The remaining equivalences are Theorem 4.3, and the last clause is the formula applied to $c \rightarrow d$. $\square$

4.2 Steps, covers, and cells

A square s is a **step** of $x \in \mathcal{R}(d)$ when $\text{Cr}(x) \triangle \{s\}$ is biclosed; the run reversing that set is the **flip** of x at s , written x^s .

Lemma 4.5. A square is a step of x if and only if its two events are consecutive for $<_x$, and x has exactly $\deg d$ steps.

Proof. A block's events are consecutive for $<_x$ (Lemma 4.2), so “consecutive in the block” and “consecutive overall” agree. If the two events of s are consecutive, toggling s swaps an adjacent pair of a linear order, so the result is again a linear order and is biclosed by Theorem 4.3. If a third event of the block fires strictly between them, toggling s swaps the pair but not the order of either event against that third one, which would then have to lie both before and after the pair.

So the steps are the covers of $<_x$ lying inside a block, and $<_x$ is a chain on N elements and blk is a monotone surjection from it onto the r blocks (Lemma 4.2); such a map collapses exactly $N - r = \deg d$ covers. $\square$

Lemma 4.6. If $x < z$ in $\mathcal{R}(d)$ then $\text{Cr}(z)$ contains a step of x that $\text{Cr}(x)$ does not. Consequently $x < y$ exactly when $\text{Cr}(y) = \text{Cr}(x) \cup \{s\}$ for such a step s , and then $y = x^s$. So $|\text{Cr}|$ grades $\mathcal{R}(d)$, from 0 at $\perp_d$ to $|\text{Sq}(d)|$ at $\top_d$.

Proof. If z agreed with x on every pair covering for $<_x$, then $<_x$ would be contained in $<_z$ by transitivity, and two linear orders so related are equal, so $z = x$. So some covering pair $p <_x q$ has $q <_z p$. Its two events share a block, since otherwise Lemma 4.2 would make every run fire p first, so the pair is a step s of x (Lemma 4.5). If $s = (p, q)$ then $s \in \text{Cr}(z) \setminus \text{Cr}(x)$, as wanted; if $s = (q, p)$ then $s \in \text{Cr}(x) \subseteq \text{Cr}(z)$, which says $p <_z q$, a contradiction.

Given such an s we have $x \leq x^s \leq y$, forcing $y = x^s$; conversely two biclosed sets differing by one element are a covering pair, by Theorem 4.3. $\square$

Say that a set S of steps of x **risers** from x when $S \cap \text{Cr}(x) = \emptyset$.

Proposition 4.7. *Let $x \in \mathcal{R}(d)$. Sending a factorisation $X \rightarrow m \rightarrow d$ of x to the set of steps of x whose two events share a block of m is a bijection onto the sets of steps of x . Write m_S for the middle it sends to S , and $\iota_S : X \rightarrow m_S$, $\pi_S : m_S \rightarrow d$ for the two legs. Then $\deg m_S = |S|$, and ι_S is order preserving whenever S rises from x .*

Proof. A middle m makes the N events into its blocks, which by Lemma 4.2 applied to ι are intervals of $<_x$, and π requires each such interval to lie inside one block of d ; conversely such an interval partition arises from the chain whose blocks are its parts, in order. The factorisation is determined by that shape: a map of wedges is determined by its action on events and injective there (Proposition 2.1), hence a monomorphism, while the first leg is forced by which events fall in which part. An interval partition of a chain is recorded by the set of covers it does not cut; those covers lie inside a block, so they are steps (Lemma 4.5); and $\deg m$ is N minus the number of parts, which is the number of uncut covers.

Let S rise from x . Since ι_S is a run over m_S , its crossings are the squares of m_S that x reverses, that is the pairs inside one part that x reverses. The covers of $<_x$ inside a part all lie in S , hence none is reversed, so $<_x$ agrees with the event order on consecutive pairs of the part and hence, by transitivity, on every pair of it. So $\text{Cr}(\iota_S) = \emptyset$. $\square$

Definition 4.8. For runs X, Y of K , an n -cell from X to Y is a chain e with $\deg e = n$, $\text{dom } \perp_e = X$ and $\text{dom } \top_e = Y$.

► *One sentence: so the n -cells are the chains of degree n , sorted by the two runs each spans, and a 0-cell is a run.*

Lemma 4.9. *Let S rise from $x \in \mathcal{R}(d)$. Then*

$$\text{Cr}(\pi_S \top_{m_S}) = \text{Cr}(x) \cup \pi_{S*}(\text{Sq}(m_S)),$$

and every $z \geq x$ with $S \subseteq \text{Cr}(z)$ satisfies $\pi_{S}(\text{Sq}(m_S)) \subseteq \text{Cr}(z)$. So $\pi_S \top_{m_S}$ is the least element of $\mathcal{R}(d)$ above x reversing all of S .*

Proof. Proposition 4.4 gives $\text{Cr}(\pi_S \top_{m_S}) = \pi_{S*} \text{Sq}(m_S) \cup \text{Cr}(\pi_S)$, and Lemma 3.4 applied to $\pi_S \iota_S = x$ with $\text{Cr}(\iota_S) = \emptyset$ gives $\text{Cr}(\pi_S) = \text{Cr}(x)$.

For the second claim list a part as $p_1 < \dots < p_k$; as in the previous proof this is also the restriction of $<_z$ to it, so each (p_i, p_{i+1}) lies in S and is reversed by z . Since $\text{Cr}(z)$ is biclosed, the first clause applied to the triples (p_i, p_{i+1}, p_j) gives, by induction on $j - i$, that z reverses every pair of the part. $\square$

Corollary 4.10. *Let $x \leq y$ in $\mathcal{R}(d)$, with step s . Then $\gamma(x \leq y) := m_{\{s\}}$ is a 1-cell from $\text{dom } x$ to $\text{dom } y$, with $\pi_{\{s\}} \perp_{m_{\{s\}}} = x$ and $\pi_{\{s\}} \top_{m_{\{s\}}} = y$; and every 1-cell α is γ of the unique cover $\perp_\alpha \leq \top_\alpha$ of $\mathcal{R}(\alpha)$. So γ , together with $x \mapsto \text{dom } x$ on vertices, is a map of quivers from $\text{Hasse}(\mathcal{R}(d))$ to the quiver of 1-cells, and $\gamma(qe) = \gamma(e)$ for every refinement $q : o \rightarrow d$ and every cover e of $\mathcal{R}(o)$.*

Proof. $m_{\{s\}}$ has degree 1 and $\text{dom } \perp_{m_{\{s\}}} = \text{dom } x$ by Proposition 4.7; by Lemma 4.9 the image of its top is the least element above x that reverses s , which is $x^s = y$. Conversely a chain α of degree 1 has one square, so $\mathcal{R}(\alpha)$ is the two-element poset $\perp_\alpha \leq \top_\alpha$, and the factorisation through its one step is α . For the last claim the step of qe is q_* of the step of e (Proposition 4.4), and q_* carries $<_x$ to $<_{qx}$, so the two factorisations cut the same partition and have the same middle. $\square$

4.3 The attachment of a 2-cell

Lemma 4.11. *Order $\text{Sq}(d)$ lexicographically, by the smaller event and then the larger. Every lower set of that order is biclosed, and so is every upper set. Consequently, for $\prec$ either that order or its reverse, the runs x_t with $\text{Cr}(x_t)$ the first t squares form a maximal chain $\perp_d = x_0 \prec \dots \prec x_{|\text{Sq}(d)|} = \top_d$, in which the $(t+1)$ -st square is a step of x_t .*

Proof. On a triple $p < q < r$ of one block the three squares come in the order $(p, q) \prec (p, r) \prec (q, r)$. For a lower set R , membership of (q, r) forces that of (p, r) , and membership of (p, r) forces that of (p, q) , which are the two clauses; for an upper set, $(p, q) \in R$ forces $(p, r) \in R$ and $(p, r) \in R$ forces $(q, r) \in R$. The chain then exists by Theorem 4.3, consecutive terms differ by one element, and Lemma 4.6 applies. $\square$

Proposition 4.12. *Let $\deg e = 2$, and write $\text{src}(e)$ and $\text{tgt}(e)$ for the words of 1-cells that γ reads off the two chains of Lemma 4.11, for the lexicographic order and for its reverse. Then $\mathcal{R}(e)$ is a polygon on $2|\text{Sq}(e)|$ elements whose only two maximal chains from $\perp_e$ to $\top_e$ are those of Lemma 4.11; so $\text{src}(e)$ and $\text{tgt}(e)$ have length $|\text{Sq}(e)|$ and differ in their first letter. Moreover either e has one block of dimension 3, $|\text{Sq}(e)| = 3$ and $\mathcal{R}(e)$ is a hexagon, or e has two blocks of dimension 2, $|\text{Sq}(e)| = 2$ and $\mathcal{R}(e)$ is a square.*

Proof. Every $x \in \mathcal{R}(e)$ has exactly two steps (Lemma 4.5), and the two flips are its two neighbours in $\text{Hasse}(\mathcal{R}(e))$, upward or downward according to whether the step is already reversed (Lemma 4.6). So $\text{Hasse}(\mathcal{R}(e))$ is 2-regular and therefore a disjoint union of cycles; $|\text{Cr}|$ rises along a directed path, so such a path leaves each vertex by the edge it did not arrive on, while both edges at $\perp_e$ point up and both at $\top_e$ point down. The cycle through $\perp_e$ thus meets $\top_e$ and splits into two directed arcs of equal length, and it exhausts $\mathcal{R}(e)$ because every run lies between $\perp_e$ and $\top_e$. Those arcs are the two chains of Lemma 4.11, which leave $\perp_e$ by different covers since the least and the greatest square differ. Finally $\deg e = 2$ forces the block dimensions above 1 to be $\{3\}$ or $\{2, 2\}$, and Definition 3.1 counts $\binom{3}{2} = 3$ and $2\binom{2}{2} = 2$. $\square$

The two species are drawn in Figure 1 and Figure 2.

Definition 4.13. The **Artin polygraph** $\text{Art}(K)$ has the runs of K as 0-cells. Its 1-cells and 2-cells from X to Y are the 1-cells and 2-cells from X to Y of Definition 4.8, that is the chains of degree 1 and 2 spanning those two runs, and each 2-cell e is attached along the pair of words $\text{src}(e)$, $\text{tgt}(e)$ of Proposition 4.12.

► *One paragraph on the name: $\text{Art}(K)$ is built from K alone, with no braid theory in it, and Proposition 4.12 already exhibits its 2-cells as braid and commutation relations; Theorem 7.5 is what makes the name literal.*

Proposition 4.14. *Read in $\langle \text{Art}(K) \rangle$, the map of quivers γ satisfies the hypothesis of Theorem 2.4 on $\mathcal{R}(d)$. So it extends uniquely to a functor $\Gamma_d : \mathcal{R}(d) \rightarrow \langle \text{Art}(K) \rangle$, and $\Gamma_d(qx) = \Gamma_o(x)$ for every refinement $q : o \rightarrow d$ and every $x \in \mathcal{R}(o)$.*

Proof. Let $x < y_1$, $x < y_2$ be distinct covers with steps $s_1 \neq s_2$, and z a common upper bound. Then $S = \{s_1, s_2\}$ rises from x , so m_S is a chain of degree 2, that is a 2-cell (Proposition 4.7), and $j := \pi_S \top_{m_S}$ satisfies $j \leq z$ (Lemma 4.9). By Proposition 4.12 the two maximal chains of $\mathcal{R}(m_S)$ are $\text{src}(m_S)$ and $\text{tgt}(m_S)$, and they leave $\perp_{m_S}$ by its two covers. Composition with π_S carries $\perp_{m_S}$ to x and $\top_{m_S}$ to j , carries covers to covers (Proposition 4.4, Lemma 4.6) and keeps the 1-cell each cover reads as (Corollary 4.10). So the two chains become two paths from x to j , leaving by y_1 and by y_2 and reading as $\text{src}(m_S)$ and $\text{tgt}(m_S)$, which are equal in $\langle \text{Art}(K) \rangle$.

For the last claim, composition with q is monotone, so $x \mapsto \Gamma_d(qx)$ and Γ_o are functors out of $\mathcal{R}(o)$; they agree on covers by Corollary 4.10, hence agree, by the last clause of Theorem 2.4. $\square$

5 The presentation

► *Frame the proof as a comparison of two universal properties: a functor out of $\langle \text{Art}(K) \rangle$ is an arrow for each degree-one chain identifying the two words of every degree-two chain (Proposition 2.2), and a functor out of $\text{Br}(K)$ is a functor on $\text{Ch}(K)^{\text{op}}$ inverting W (Proposition 2.3). Each is built from the other.*

Fix K throughout this section, and write E, Θ, Φ for the functors called E_K, Θ_K, Φ_K elsewhere.

An n -cell α from X to Y comes with the two refinements $\perp_\alpha : X \rightarrow \alpha$ and $\top_\alpha : Y \rightarrow \alpha$, of which the first is order preserving; since Q inverts $\perp_\alpha^{\text{op}}$, we may set

$$E(X) := Q(X), \quad E(\alpha) := Q(\top_\alpha^{\text{op}}) \circ Q(\perp_\alpha^{\text{op}})^{-1} : Q(X) \rightarrow Q(Y).$$

Lemma 5.1. *Let d be a chain and $z \in \mathcal{R}(d)$. Every path from $\perp_d$ to z in $\text{Hasse}(\mathcal{R}(d))$ reads, under $E \circ \gamma$, as $Q(z^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1}$. The two words of a 2-cell e are both paths from $\perp_e$ to $\top_e$ (Proposition 4.12); they therefore have equal image, and E extends to a functor $\langle \text{Art}(K) \rangle \rightarrow \text{Br}(K)$ by Proposition 2.2(i).*

Proof. Induct on the length of the path, the empty path reading as the identity. For a cover $x < y$ with step s , put $\beta := \gamma(x < y)$ and $q := \pi_{\{s\}}$, so $x = q\perp_\beta$ and $y = q\top_\beta$ (Corollary 4.10) and $x^{\text{op}} = \perp_\beta^{\text{op}} q^{\text{op}}$. The middle two factors of

$$E(\beta) \circ Q(x^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1} = Q(\top_\beta^{\text{op}}) \circ Q(\perp_\beta^{\text{op}})^{-1} \circ Q(\perp_\beta^{\text{op}}) \circ Q(q^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1}$$

cancel, leaving $Q(\top_\beta^{\text{op}})Q(q^{\text{op}})Q(\perp_d^{\text{op}})^{-1} = Q(y^{\text{op}})Q(\perp_d^{\text{op}})^{-1}$. $\square$

Lemma 5.2. *Setting $\Theta(d) := \text{dom } \perp_d$ and, for a refinement $u : c \rightarrow d$,*

$$\Theta(u^{\text{op}}) := \Gamma_d(\perp_d \leq u\perp_c) : \text{dom } \perp_d \rightarrow \text{dom } \perp_c$$

— which typechecks because $u \perp_c \in \mathcal{R}(d)$ has source $\text{dom } \perp_c$ — defines a functor $\Theta : \text{Ch}(K)^{\text{op}} \rightarrow \langle \text{Art}(K) \rangle$; it sends u^{op} to an identity whenever u is order preserving, so it descends uniquely to $\Phi : \text{Br}(K) \rightarrow \langle \text{Art}(K) \rangle$ with $\Phi Q = \Theta$.

Proof. Let $v : b \rightarrow c$ and $u : c \rightarrow d$. Composition with u is monotone (Proposition 4.4), so $\perp_d \leq u \perp_c \leq uv \perp_b$, and Γ_d being a functor, $\Theta((uv)^{\text{op}}) = \Gamma_d(u \perp_c \leq uv \perp_b) \circ \Theta(u^{\text{op}})$; the left factor is $\Gamma_c(\perp_c \leq v \perp_b) = \Theta(v^{\text{op}})$ by the last clause of Proposition 4.14. If u is order preserving then $u \perp_c = \perp_d$ (Proposition 4.4), so the comparison is an identity. The descent is Proposition 2.3. $\square$

Theorem 5.3. *For every bipointed precubical set K , the functor E is an equivalence $\langle \text{Art}(K) \rangle \rightarrow \text{Br}(K)$, with Φ inverse to it.*

Proof. On a 0-cell, $\Phi(E(X)) = \Theta(X) = \text{dom } \perp_X = X$ (Proposition 4.4). On a 1-cell α , applying Φ to the defining composite gives $\Phi(E(\alpha)) = \Theta(\top_\alpha^{\text{op}}) \circ \Theta(\perp_\alpha^{\text{op}})^{-1}$, whose second factor is an identity because $\perp_\alpha$ is order preserving (Lemma 5.2). Since $\top_\alpha \perp_Y = \top_\alpha$ and $\mathcal{R}(\alpha)$ is the two-element poset $\perp_\alpha \leq \top_\alpha$,

$$\Theta(\top_\alpha^{\text{op}}) = \Gamma_\alpha(\perp_\alpha \leq \top_\alpha) = q(\gamma(\perp_\alpha \leq \top_\alpha)) = q(\alpha)$$

by Corollary 4.10. So ΦE agrees with the identity on 0- and 1-cells, hence is the identity (Proposition 2.2(ii)).

For the other composite, Lemma 5.1 gives $E(\Theta(u^{\text{op}})) = Q((u \perp_c)^{\text{op}}) Q(\perp_d^{\text{op}})^{-1} = Q(\perp_c^{\text{op}}) \circ Q(u^{\text{op}}) \circ Q(\perp_d^{\text{op}})^{-1}$, which is $Q(u^{\text{op}})$ conjugated by the isomorphisms $Q(c) \cong Q(\text{dom } \perp_c)$ that Q makes of the order-preserving refinements $\perp_c$. So $E\Theta \cong Q$, and $E\Phi \cong \text{id}$ by uniqueness of descent (Proposition 2.3). $\square$

► *One sentence: the comparison is an equivalence and not an isomorphism because $\text{Br}(K)$ is not skeletal — a chain and the run below it become isomorphic there without being equal — so the runs are a skeleton of $\text{Br}(K)$.*

6 Naturality

Theorem 6.1. *Sending K to $\text{Art}(K)$, and $f : K \rightarrow L$ to $\text{Ch}(f)$ on cells in every dimension, is a functor $\square_* \rightarrow \text{Poly}$; and for every $f : K \rightarrow L$,*

$$E_L \circ \langle \text{Art}(f) \rangle = \text{Br}(f) \circ E_K : \langle \text{Art}(K) \rangle \longrightarrow \text{Br}(L).$$

Proof. A run over a chain e is determined by its map of wedges (Definition 4.1), and $\text{Ch}(f)e$ has the same shape and the same maps of wedges into it, so $\mathcal{R}(e) = \mathcal{R}(\text{Ch}(f)e)$ as ordered sets. Everything in Definition 3.2, Lemma 4.5 and Corollary 4.10 is defined from the map of wedges alone, so $\text{Ch}(f)$ preserves $\perp$, $\top$, cells, steps, covers and γ ; and it carries the lexicographic order on $\text{Sq}(e)$ to the lexicographic order on $\text{Sq}(\text{Ch}(f)e) = \text{Sq}(e)$, so it preserves src and tgt letter by letter. Hence $\text{Art}(f)$ is a morphism of polygraphs, functorially in f because Ch is a functor.

Both composites send a run X to $Q(\text{Ch}(f)X)$. On a 1-cell α , $\text{Br}(f)$ is the descent of $Q_L \circ \text{Ch}(f)^{\text{op}}$, so it sends $E_K(\alpha)$ to $Q((\text{Ch}(f)\top_\alpha)^{\text{op}}) \circ Q((\text{Ch}(f)\perp_\alpha)^{\text{op}})^{-1}$. Here $\text{Ch}(f)\perp_\alpha$ is an order-preserving refinement into $\text{Ch}(f)\alpha$ out of a run, hence is $\perp_{\text{Ch}(f)\alpha}$ (Proposition 4.4), and likewise $\text{Ch}(f)\top_\alpha = \top_{\text{Ch}(f)\alpha}$. So both composites send α to $E_L(\text{Ch}(f)\alpha)$, and Proposition 2.2(ii) applies. $\square$

► *One sentence: the square is an equality, not an isomorphism, because an order-preserving refinement is determined by its source and target (Proposition 3.6), so transporting the two legs of a cell involves no choice. The tree states it as an isomorphism pinned at the identity, which is this equality read through eqToIso.*

7 The point

Since Z is terminal, every serial wedge has exactly one bipointed map to it and every map of wedges commutes over it, so $\text{Ch}(Z)$ is the category of shapes and maps of wedges. Write Z_N for the run of Z on N events. Both $\perp_e$ and $\top_e$ have source a run on $|e|$ events, so every cell of Z is a loop at $Z_{|e|}$.

For a shape a on N events put $\text{cut}(a) := \{k \in [N-1] : \text{blk}_a(k) = \text{blk}_a(k+1)\}$, and write $\text{cut}(d)$ for the cuts of a chain's shape; for $i \neq k$ in $[N-1]$ let $\text{cox}(i, k) := 3$ when $|i - k| = 1$ and $\text{cox}(i, k) := 2$ otherwise.

Lemma 7.1. *$a \mapsto \text{cut}(a)$ is a bijection from the shapes on N events onto the subsets of $[N-1]$, with $|\text{cut}(a)| = \deg a$. Hence the n -cells at Z_N are the n -element subsets of $[N-1]$; write σ_k for the 1-cell with $\text{cut} = \{k\}$. Moreover a chain e of Z with $\deg e = 2$ and $\text{cut}(e) = \{i < k\}$ has $|\text{Sq}(e)| = \text{cox}(i, k)$.*

Proof. A shape on N events is a composition of N and $\text{cut}(a)$ is the complement of its set of block boundaries in $[N-1]$, so the first claim is the classical correspondence [Sta12, §1.2] and $|\text{cut}(a)| = N - r = \deg a$. For the last claim, e has a block of dimension 3 exactly when three consecutive events share a block, that is exactly when $k = i + 1$; then $|\text{Sq}(e)| = 3$, and otherwise $|\text{Sq}(e)| = 2$, by Proposition 4.12. $\square$

For a step s of x , write $\text{pos}_x(s)$ for the earlier of the two positions $<_x$ gives the events of s ; so x passes s between positions $\text{pos}_x(s)$ and $\text{pos}_x(s) + 1$.

Lemma 7.2. *Let d be a chain on N events and $x \in \mathcal{R}(d)$. Then:*

1. pos_x is a bijection from the steps of x onto $\text{cut}(d)$;
2. $\text{pos}_{x^s}(s) = \text{pos}_x(s)$;
3. $\text{pos}_{\perp_d}(s) = p$, where $s = (p, q)$.

If d is a chain of Z , then a cover $x \triangleleft y$ with step s has $\gamma(x \triangleleft y) = \sigma_{\text{pos}_x(s)}$.

Proof. (i) The fibres of blk for $<_x$ are consecutive intervals of lengths $a_1, \dots, a_r$ (Lemma 4.2), independent of x , so the set of positions t with t and $t + 1$ in one block is $\text{cut}(d)$ for every x . A step is a cover of $<_x$ inside a block (Lemma 4.5), so pos_x lands in $\text{cut}(d)$, and it is injective because distinct covers of a chain have distinct lower ends; both sides have $\deg d$ elements (Lemma 4.5, Lemma 7.1). (ii) Flipping s exchanges its two times and changes no other. (iii) $\perp_d$ fires the events in their own order.

For the last claim, $\gamma(x \triangleleft y) = m_{\{s\}}$, whose blocks are the parts of $<_x$ cut at every cover but s (Proposition 4.7); so $m_{\{s\}}$ has one block of dimension 2, at positions $\text{pos}_x(s)$ and $\text{pos}_x(s) + 1$, and edges elsewhere, so its cut set is $\{\text{pos}_x(s)\}$. $\square$

Proposition 7.3. *Let e be a chain of Z with $\deg e = 2$ and $\text{cut}(e) = \{i < k\}$, and let $\perp_e = x_0 \triangleleft \cdots \triangleleft x_{|\text{Sq}(e)|} = \top_e$ be the chain of Lemma 4.11 for the lexicographic order, with s_t the $(t+1)$ -st square. Then $\text{pos}_{x_t}(s_t) = i$ for t even and k for t odd, so*

$$\text{src}(e) = \sigma_i \sigma_k \sigma_i \cdots, \quad \text{tgt}(e) = \sigma_k \sigma_i \sigma_k \cdots,$$

both of length $\text{cox}(i, k)$.

Proof. Every x_t has exactly two steps, passed at i and at k (Lemma 7.2(i)).

For $t = 0$: $\perp_e$ passes (p, q) at p (Lemma 7.2(iii)), and the lexicographic order sorts $\text{Sq}(e)$ by the smaller event first. If e is one block of dimension 3 on events $i, i+1, i+2$, its squares in order are $(i, i+1)$, $(i, i+2)$, $(i+1, i+2)$, with smaller events $i, i, i+1 = k$; if e is two blocks of dimension 2 on $\{i, i+1\}$ and $\{k, k+1\}$, they are $(i, i+1)$ and $(k, k+1)$, with smaller events i and k . Either way the least square is passed at i and the greatest at k .

For the step, s_t and s_{t+1} are both steps of x_{t+1} : the first because a flip of a step is a step, the second by Lemma 4.11. Now $\text{pos}_{x_{t+1}}(s_t) = \text{pos}_{x_t}(s_t)$ (Lemma 7.2(ii)), so $\text{pos}_{x_{t+1}}(s_{t+1})$ is whichever of i and k that is not. The words follow by the last clause of Lemma 7.2, and their length is $|\text{Sq}(e)| = \text{cox}(i, k)$ (Lemma 7.1). The reverse order has the lexicographic order's greatest square first, passed at k , and the same induction applies. $\square$

Definition 7.4. **Artin's braid polygraph** $\mathcal{A}$ has one 0-cell for each $N \in \mathbb{N}$; at N , the loops $\sigma_1, \dots, \sigma_{N-1}$; and, for each pair $i < k$ in $[N-1]$, one 2-cell with words $\sigma_i \sigma_k \sigma_i \cdots$ and $\sigma_k \sigma_i \sigma_k \cdots$, both of length $\text{cox}(i, k)$.

► *One or two sentences: unwind Definition 7.4 to Artin's relations [Art47], so $\langle \mathcal{A} \rangle$ has endomorphism monoid $(B_N^+)^{\text{op}}$ at N and no arrows between distinct objects, i.e. $\langle \mathcal{A} \rangle = \mathbf{Braid}^{+, \text{op}}$.*

Theorem 7.5. $\text{Art}(Z) \cong \mathcal{A}$ in **Poly**.

Proof. The runs of Z are the Z_N and every cell of Z is a loop, so both polygraphs are disjoint unions over N and it suffices to compare at each N . There the 1-cells are indexed by $[N-1]$ and the 2-cells by the pairs $i < k$ (Lemma 7.1), and by Proposition 7.3 the words of the 2-cell at $\{i < k\}$ are those of Definition 7.4. The resulting morphism of polygraphs is bijective in every dimension, hence an isomorphism (Proposition 2.2(iii)). $\square$

Corollary 7.6. $\text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$.

Proof. Theorem 5.3 at $K = Z$ and Theorem 7.5. $\square$

► *Closing paragraph: combining with Theorem 6.1, the terminal map gives $\text{Br}(!_K) : \text{Br}(K) \rightarrow \text{Br}(Z)$, computed cell by cell — a degree-one chain goes to σ_{pos} for the position at which its run passes its one step, a degree-two chain to the relation of its two positions. Say what this sharpens in [PZ21]. Note that Matsumoto's theorem plays no role, B_N^+ being taken here as Artin's presentation.*

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